

Development of a Mathematical Model for the Simulation of Ex Vivo Ocular Tissue Clearing

Proof of “Scattering coefficient changes from Lipid Concentration”

Equation

Proof. Since the light scattering property of tissue is directly proportional to the concentration of lipids, the scattering coefficient of a layer can be based upon the percentage of remaining lipid during delipidation. The equation is as follows below:

$$\mu'_s = \mu_s \left(\frac{C_{\text{lipid}}}{C_{\text{lipid},0}} \right) + \mu_{\min}.$$

$\mu_{\min}$ was added to account for the intrinsic scattering coefficient of CUBIC and represents a functionally negligible value which can be verified using experimental data. The component μ_s is based on the scattering coefficient of each individual tissue layer prior to the delipidation procedure. While calculating the scattering coefficient of an individual tissue layer, the equation can be simplified to the following:

$$\mu_s = C_{\text{lipid},total} + \mu_{\min}$$

Alternatively, for calculations involving the scattering coefficient of the eye as a whole, a modified Mie theory equation can be used, following experimentally derived data [1].

$$\mu_s = k \left(\frac{\nu_s}{\nu_b} - 1 \right)^{2.09}.$$

The equation includes the refractive indices of the scattering medium (ν_s) and of the background fluid (ν_b). The value of k and exponent was experimentally derived to be 739.8 and 2.09, respectively [1]. Overall, this equation uses the difference in refractive index between the background and scatterer to determine the scattering coefficient of a material.

REFERENCES

- [1] T. Yu, J. Zhu, D. Li, and D. Zhu, “Physical and chemical mechanisms of tissue optical clearing,” *iScience*, vol. 24, p. 102178, 2021.